Engine

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Engines and fuel system; Four stroke working petrol engines or diesel engines model

What it is used for

An engine training model demonstrates how a petrol or diesel engine converts fuel energy into mechanical power and how its fuel system operates. Correct inspection, controlled starting, and safe observation help learners understand engine components and the four-stroke cycle without unnecessary risk.

Main parts

- Cylinder block houses the cylinders and major engine components.
- Cylinder head contains valves and combustion-chamber components.
- Piston and connecting rod transfer combustion force to the crankshaft.
- Crankshaft converts reciprocating motion into rotation.
- Camshaft operates valves in a four-stroke engine.
- Intake and exhaust systems manage airflow and exhaust gases.
- Fuel system supplies petrol or diesel to the engine.
- Cooling and lubrication systems help control temperature and friction.
- Starter, battery, and ignition components support starting where fitted.
- Training controls, gauges, and cutaway panels show operation and measurements.

Before you start

1. Only trained instructors or authorised technicians should operate the engine model.
2. Identify whether the model is petrol or diesel and review its operating manual.
3. Inspect the engine mount, guards, fuel lines, wiring, battery connections, and controls for damage or leakage.
4. Confirm that the area is well ventilated and exhaust extraction is available for a running engine.
5. Check engine oil, coolant, and fuel levels according to the manual.
6. Ensure the transmission or output shaft is in the specified safe state and the work area is clear.
7. Keep a suitable fire extinguisher accessible and remove ignition sources from fuel-handling areas.

How to use

1. Explain the main engine components and identify the intake, compression, power, and exhaust strokes.
2. Confirm that all guards, connections, fluid levels, and safety devices are ready for operation.
3. For a petrol model, follow the specified ignition and starting procedure; for a diesel model, follow its prescribed starting and preheating procedure where fitted.

4. Start the engine from the designated operating position and observe the initial gauges and warning indicators.
5. Allow the engine to run at the approved idle speed while checking for unusual noise, vibration, smoke, or leakage.
6. Use the training controls or cutaway model to observe the four-stroke sequence and fuel-system operation.
7. Record readings such as engine speed, temperature, pressure, or fuel-system measurements where instruments are provided.
8. Change operating settings only within the instructor's approved limits and according to the manual.
9. Stop the engine immediately if a warning indicator, fuel leak, abnormal noise, or unsafe condition appears.
10. Shut down using the prescribed procedure and wait for hot and moving parts to become safe before inspection.
11. Record observations, report faults, and restore the training area after the session.

⚠ Safety

- Never run an engine in an enclosed or poorly ventilated area because exhaust gases can be fatal.
- Keep hands, clothing, hair, and tools away from belts, pulleys, fans, and rotating shafts.
- Do not smoke or create sparks near fuel, fuel vapour, or open fuel-system components.
- Do not touch the exhaust manifold or other hot parts during or immediately after operation.
- Never open a hot cooling system or remove a pressurised cap.
- Isolate the battery and other energy sources as specified before electrical or mechanical servicing.

Care & storage

- Check oil, coolant, fuel lines, filters, and visible connections at the scheduled interval.
- Clean the training model and keep air intakes and cooling passages unobstructed.
- Inspect belts, hoses, wiring, mounts, guards, and battery terminals regularly.
- Use the specified fluids and approved service procedures for the engine type.
- Arrange qualified maintenance for leaks, abnormal smoke, starting faults, or inconsistent readings.

Common mistakes to avoid

- Running the engine without adequate ventilation or exhaust extraction.
- Confusing petrol-engine ignition procedures with diesel-engine starting procedures.
- Touching moving or hot components during inspection.
- Ignoring fluid leaks, warning indicators, or abnormal engine noise.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.